Pre-Lab 6: Radiometer

Submit one Word or PDF file on Canvas before your own Lab 6 section begins. Include your name and section. Run the supplied notebook, export the two figures, and answer Q1-Q5. Do not submit only code or screenshots of an application window.

What the experiment measures

A lamp heats the radiometer vanes. Gas-mediated thermal forces drive rotation, while losses resist it. You will measure passages in a video and compare speed across illumination conditions. This pre-lab uses explicitly synthetic examples, not laboratory data.

Q1. From frames to rpm

Run the passage example. Consecutive events occur at frames 0, 250, 500, 750, ... , 2000. Actual capture rate is 1000 fps. A full revolution contains four of these verified events. Calculate elapsed time, revolutions, mean rpm, and angular speed [rad/s]. Show the calculation and export the cumulative-revolutions figure.

Q2. Two common measurement mistakes

Recalculate Q1 if someone uses the movie playback rate of 60 fps. Then recalculate using p = 1 while keeping the true 1000 fps. Explain the direction and factor of each error. Explain how you will verify p in your own movie.

Q3. A simple rotational model

The notebook plots an ideal constant-torque model with final speed 90 rpm and time constants T = 1, 3, 6 s. Export the speed-versus-time figure. Which response reaches its final speed more slowly? At t = T, what fraction of the final speed has been reached from rest? Show how the exponential solution follows from the linear differential equation.

Variables and assumptions

$$I\frac{d\omega}{dt} = \tau_{\text{gas}} - \tau_{\text{loss}}$$

I = rotor moment of inertia [kg·m²]; ω = angular speed [rad/s]; τ_{gas} = thermally driven gas torque [N·m]; τ_{loss} = resisting torque [N·m]; t = time [s]. For the illustrative approximation $\tau_{loss} = b\omega$, b is the rotational damping coefficient [N·m·s]. The mechanical time constant is $T_m = I/b$ [s].

$$\omega(t) = \omega_\infty + (\omega_0 - \omega_\infty)e^{-t/T_m}$$

ω_0 = initial angular speed; ω_∞ = final steady angular speed. This solution assumes constant gas torque, constant b, and no additional dry-friction threshold. Real vane temperatures can change after the lamp turns on, so the model does not uniquely predict the experimental transient.

Q4. Physics and limits

Explain why a Crookes radiometer needs residual gas to demonstrate its usual thermally driven rotation. Is increased light always equivalent to a known increase in torque? Can two different lamp spectra be compared using lux alone as if it were W/m²? Answer briefly, using the lecture.

Q5. Your measurement plan

List what you will hold fixed, what you will vary, and what you will record for each run. Explain how you will identify a steady interval and check a missed crossing. Describe one industrial measurement technique that uses the same passage-timing idea.

Submission checklist

Q1 worked calculation and figure; Q2 error factors and p check; Q3 response figure and derivation; Q4 physical explanation; Q5 concise test plan. Use only equations you actually discuss. No coding changes are required beyond running the supplied cells.